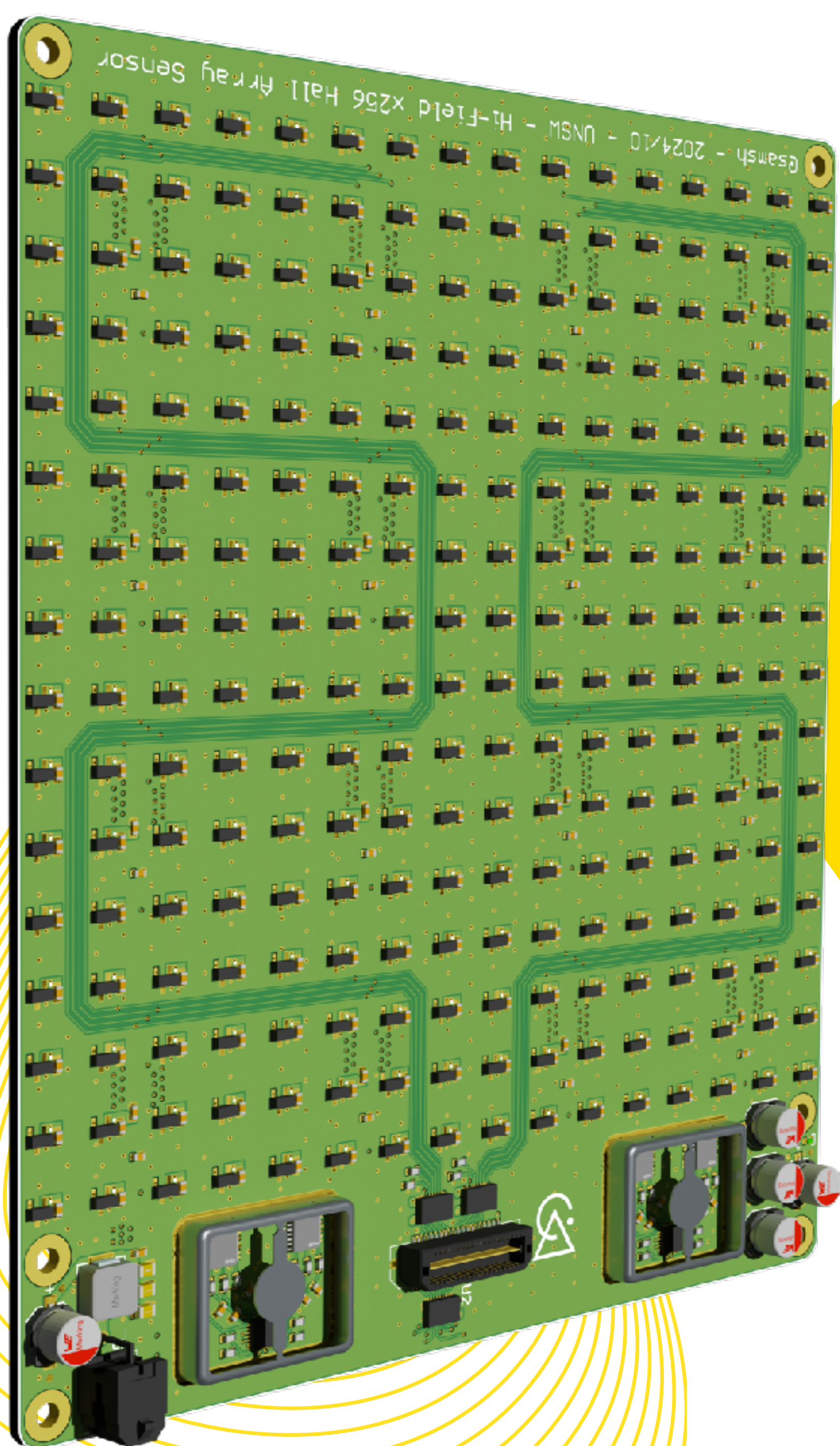


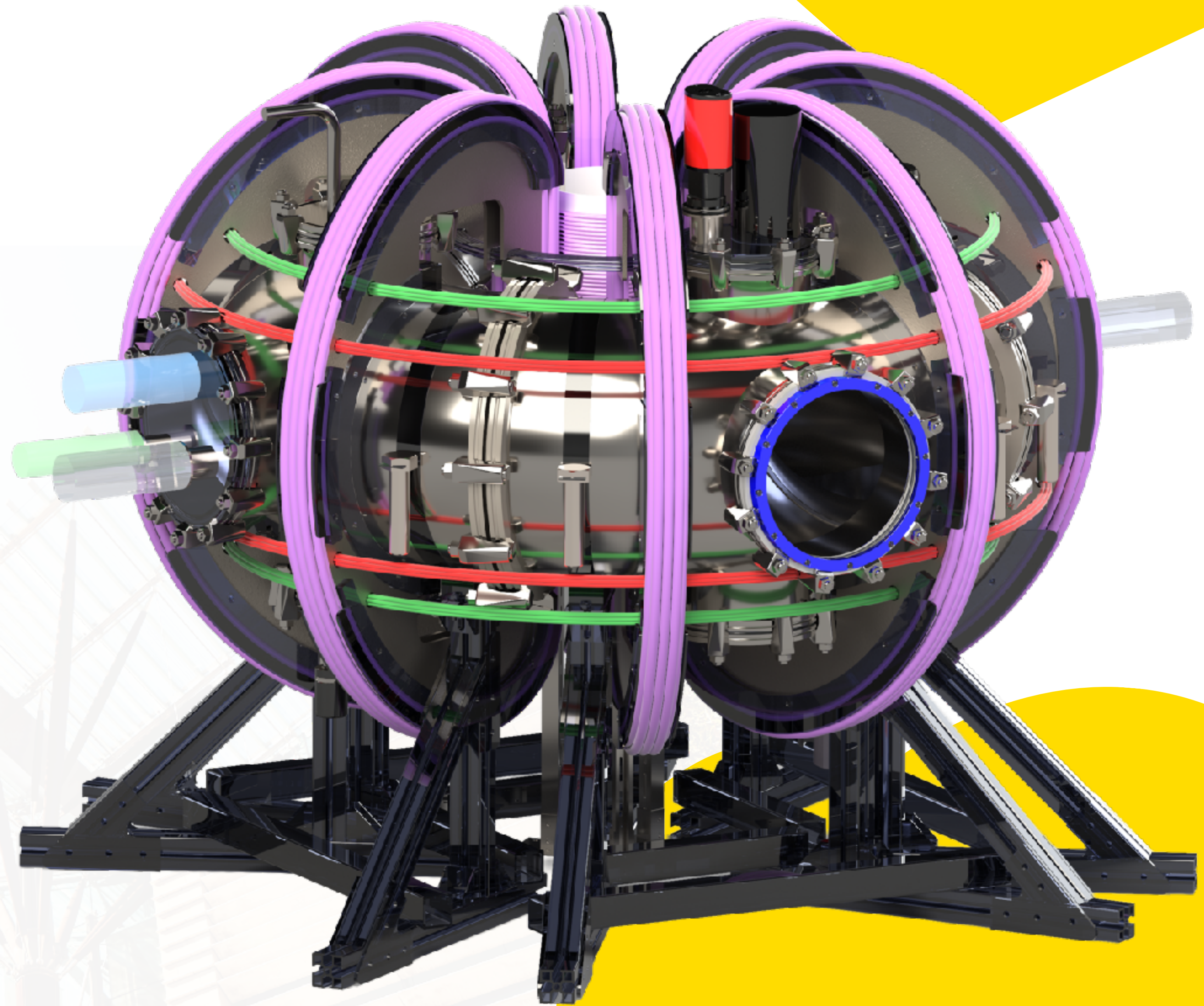


Hall Arrays for 2D Tokamak Magnetic Field Imaging

Sam Shelton, Thesis C Conference
August 2025

The SOUTH Tokamak

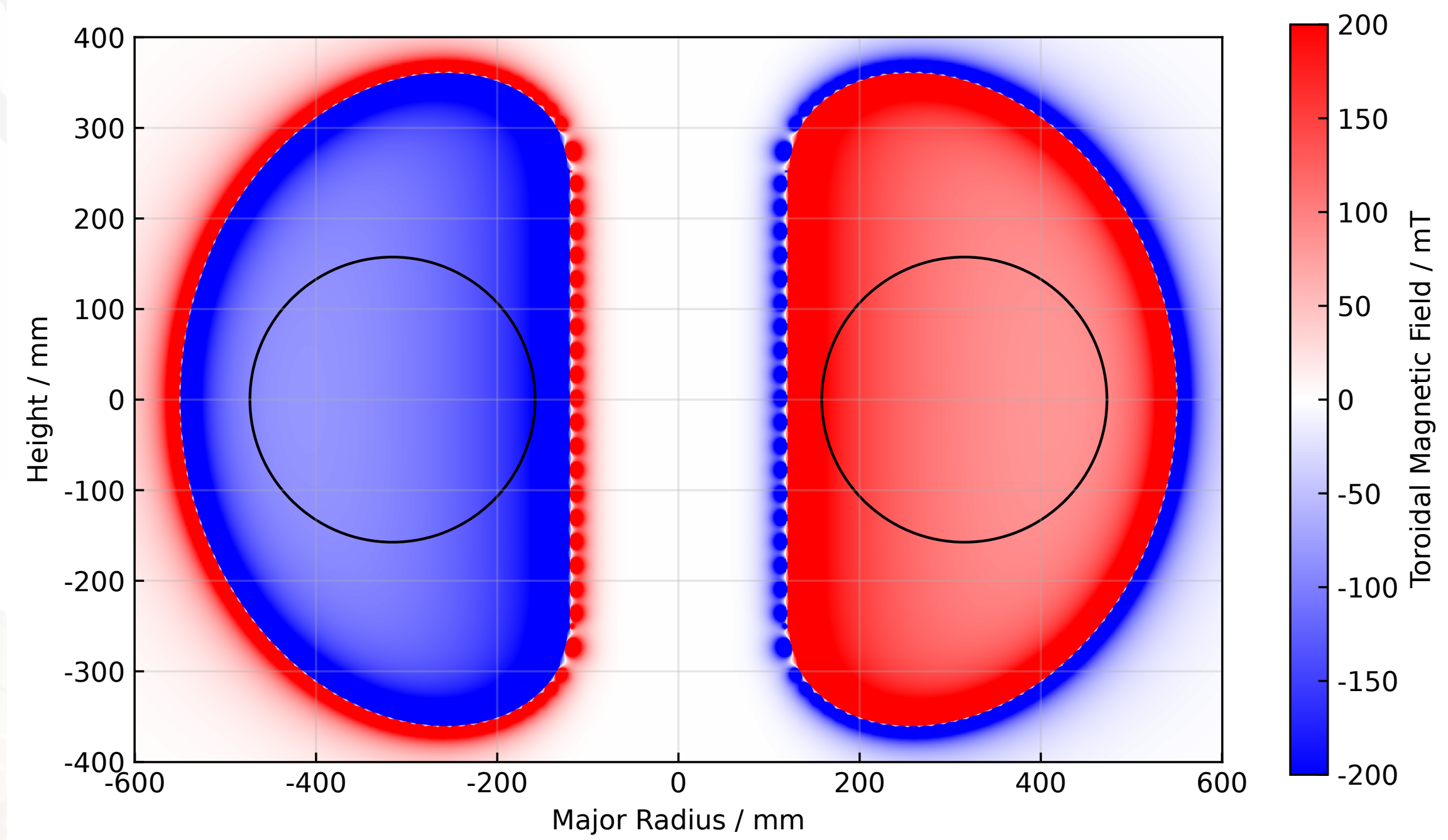
Constructed as part of the ATOMCRAFT vertically integrated project, the Student Operated Undergraduate Tokamak for Hydrogen (SOUTH) will be the worlds first fusion device of its class designed and built by undergraduates when it is switch on in late 2026.



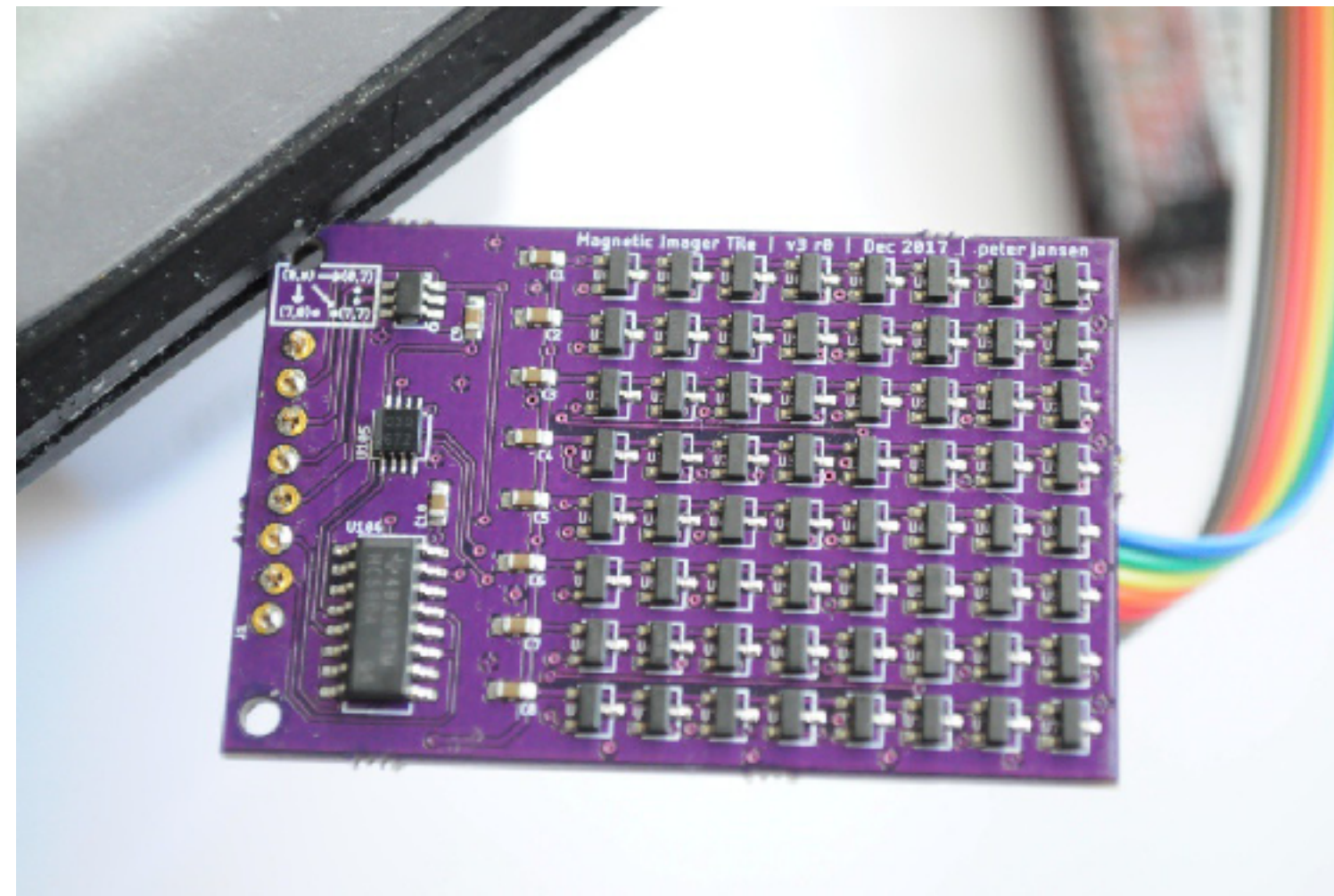


Aims

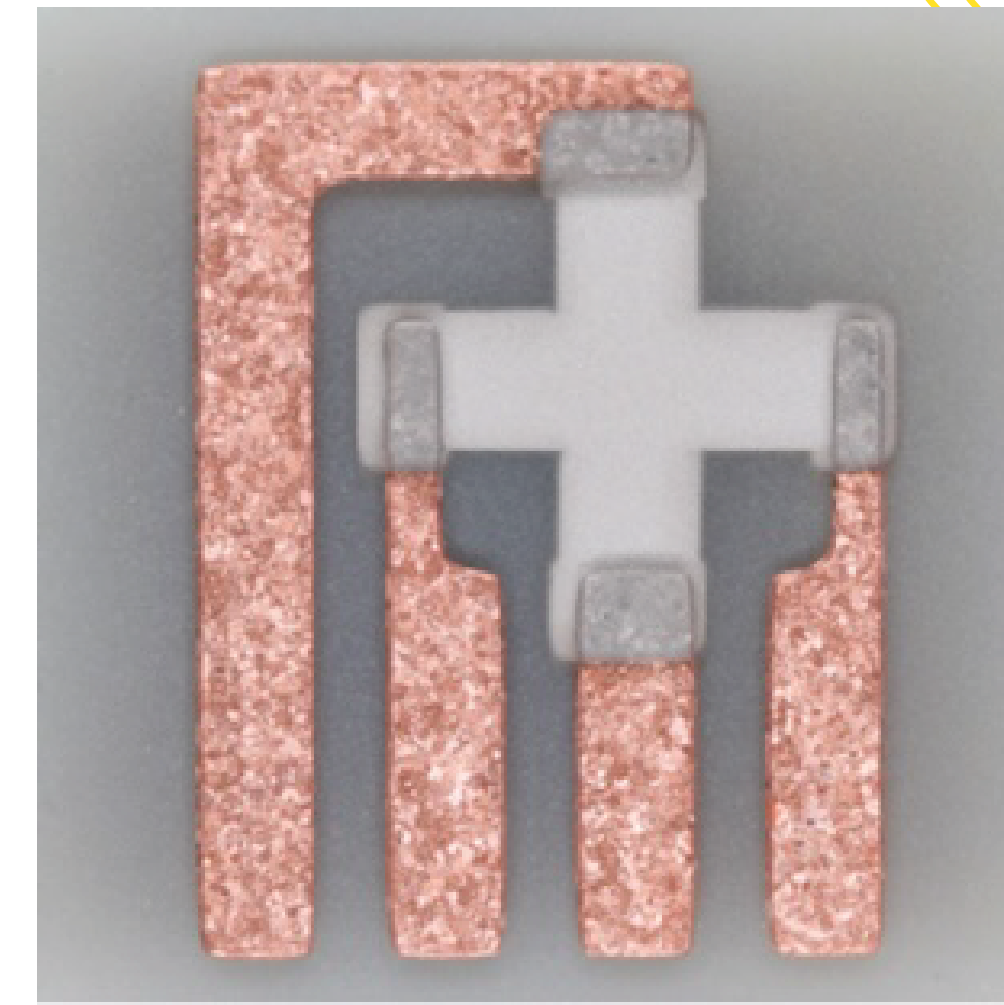
- We need a device to **monitor the toroidal field** to make sure SOUTH will actually work
- **De-risk** the R&D and manufacturing process



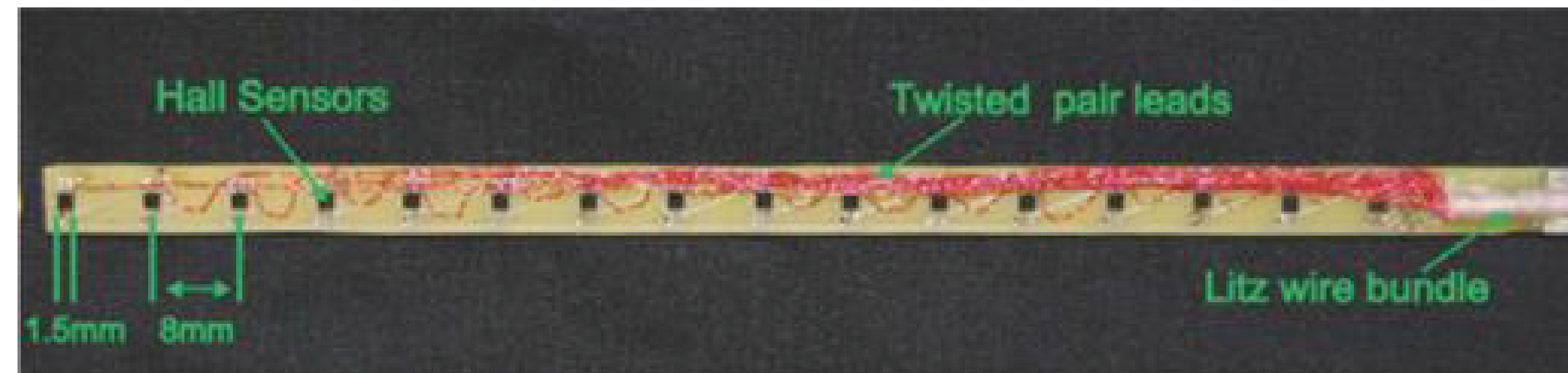
Existing Solutions



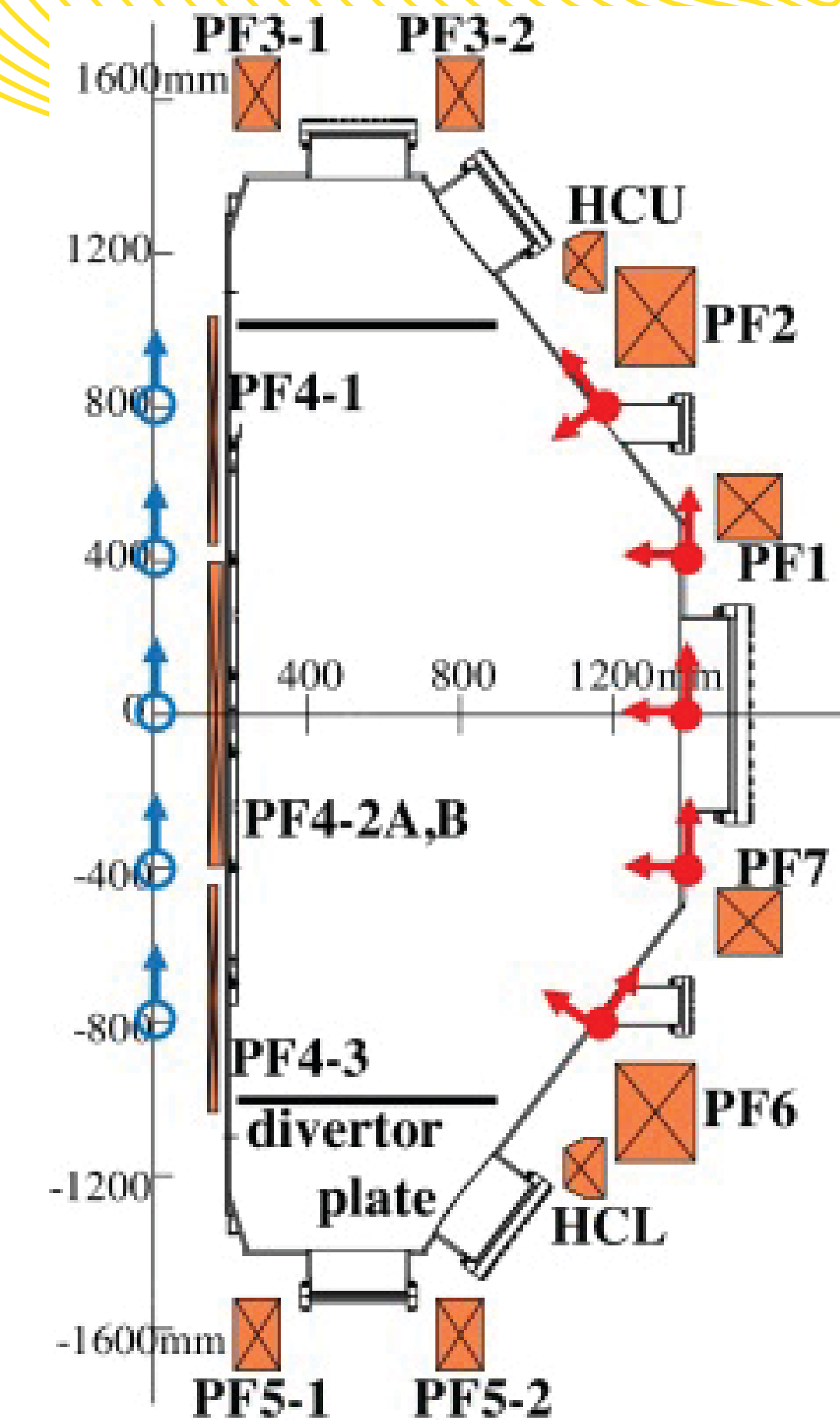
*Jansen*¹: educational tool, too slow and no filtering or buffering



*Dûran*²: single, radiation hardened sensor for ITER



*Stevenson*³: Insertable 1D poloidal array with commercial sensors

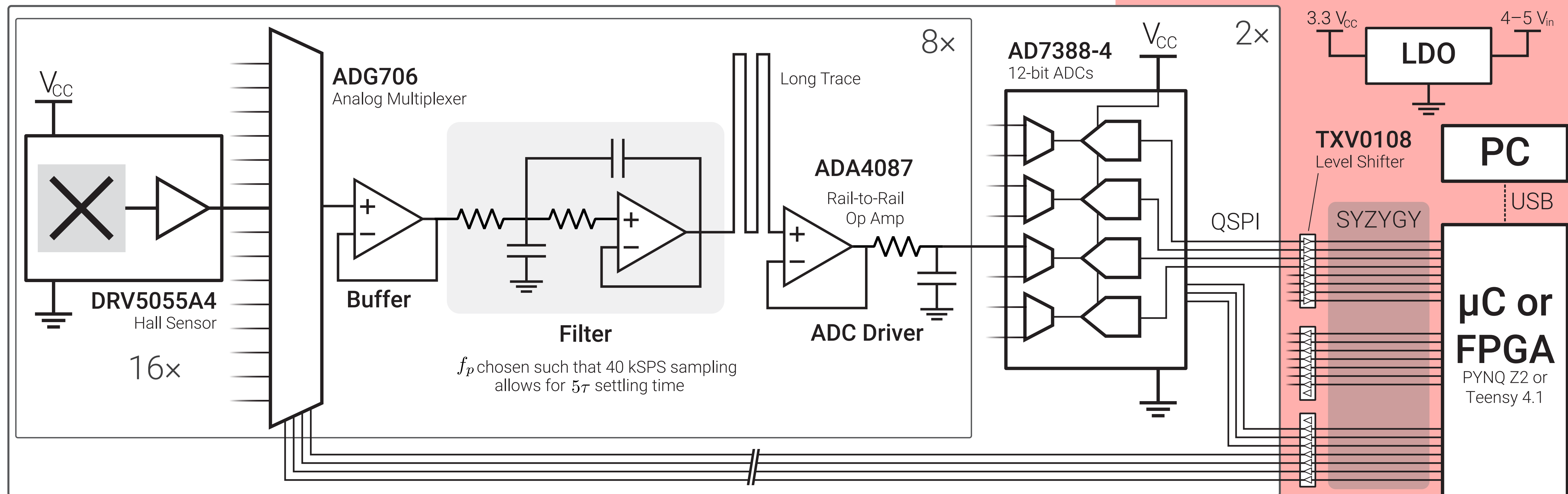


*Hasegawa*⁴: QUEST probe locations

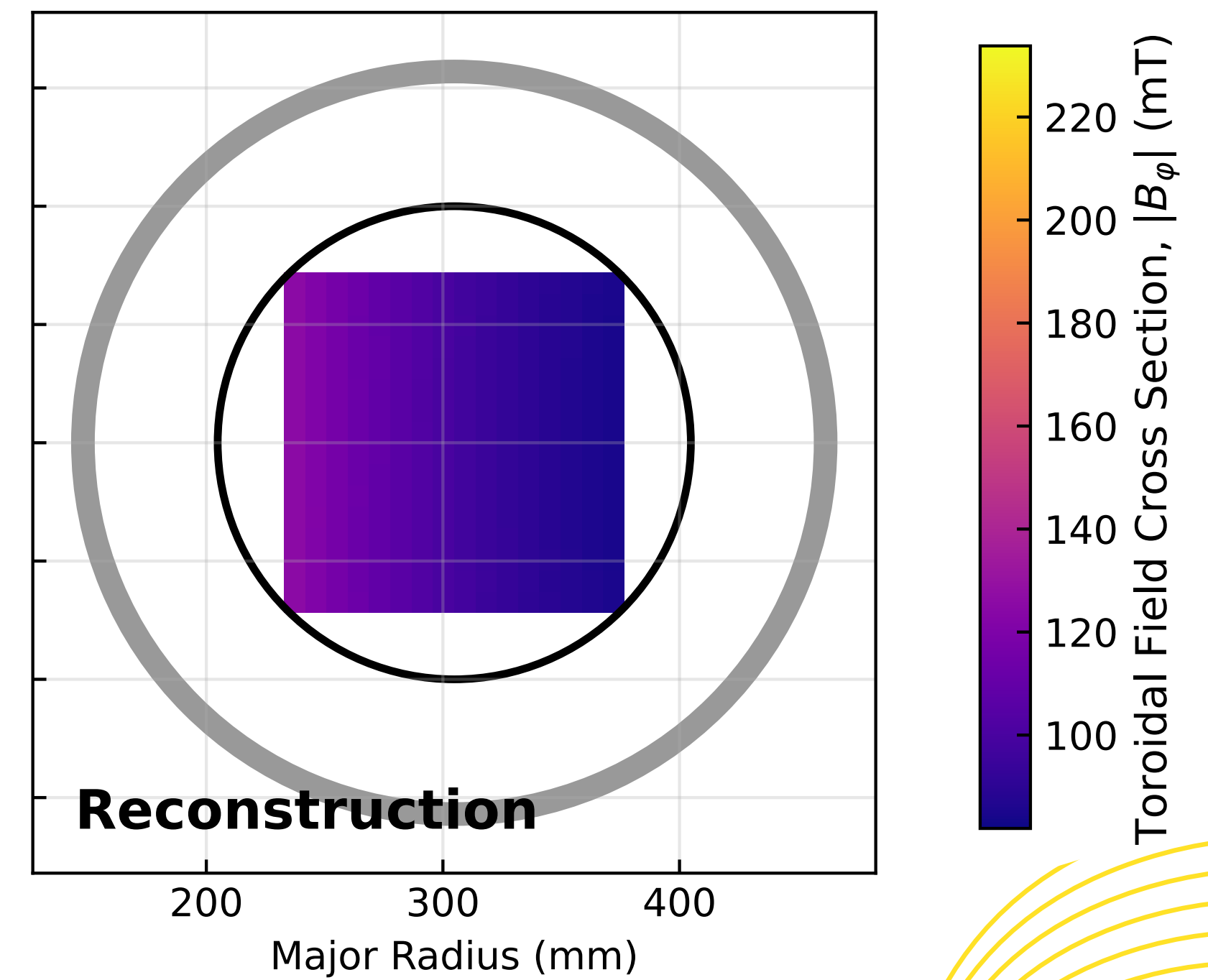
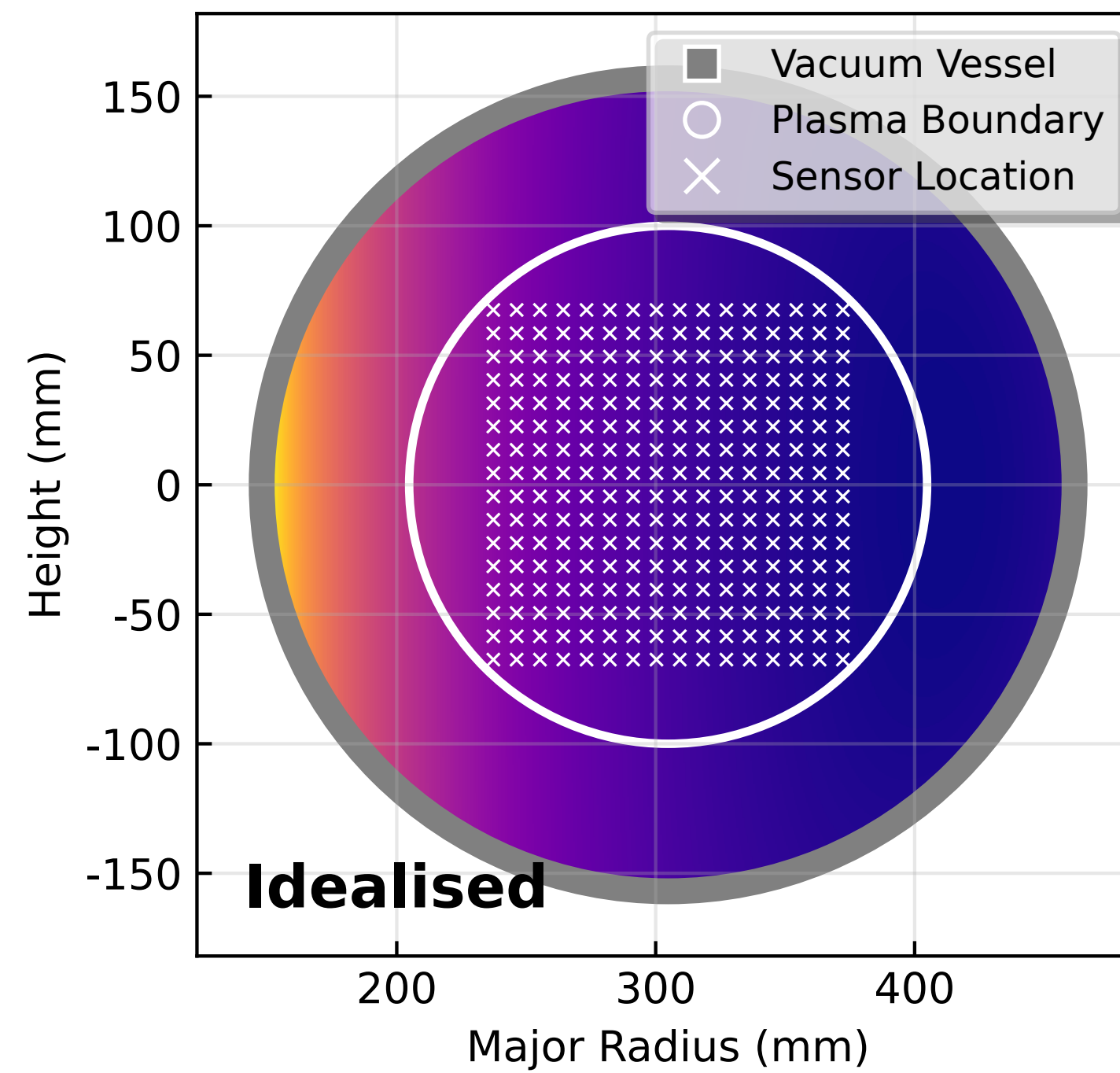
- ## Design Requirements
- High **Bandwidth**
 - High **Sensitivity & EMI Resistance**
 - Wide **Dynamic Range**
 - Must fit inside **SOUTH**

Device Methodology

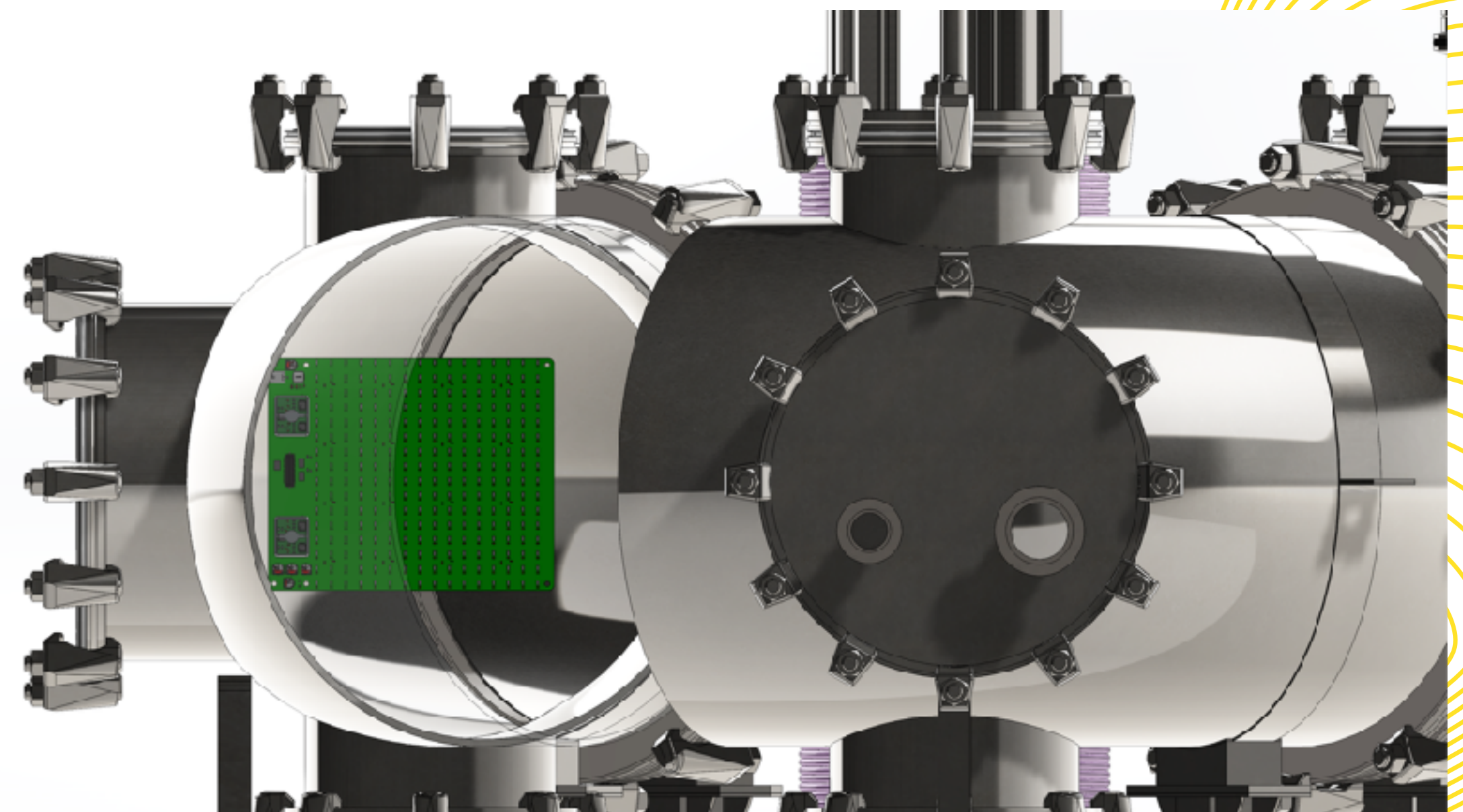
- **Paralleled ADC pipeline**
- **Multiplexing sensors**
- Comprehensive **filtering regime**



Device Methodology



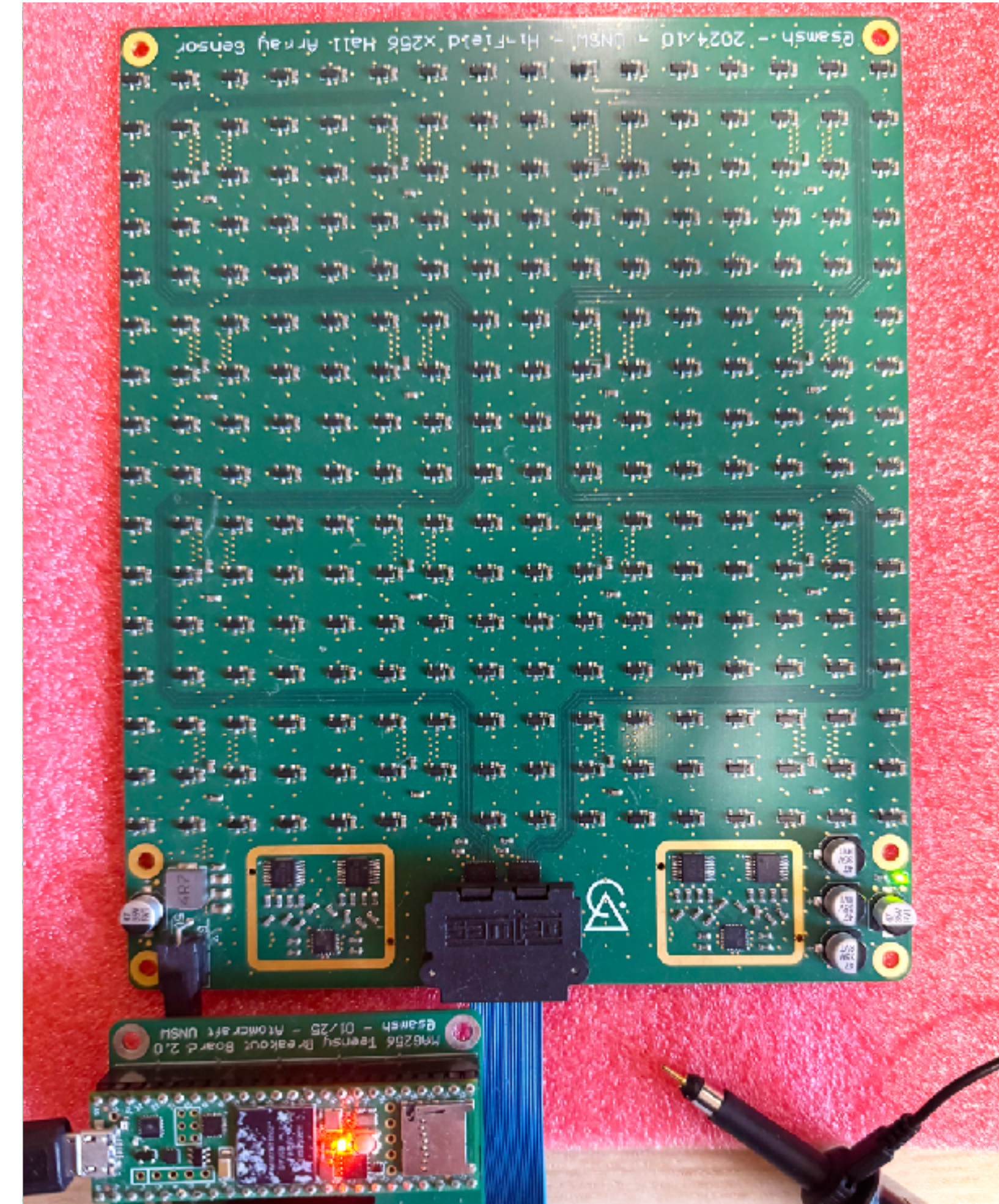
A simulated
**toroidal field
reconstruction**
and experimental
configuration



Results

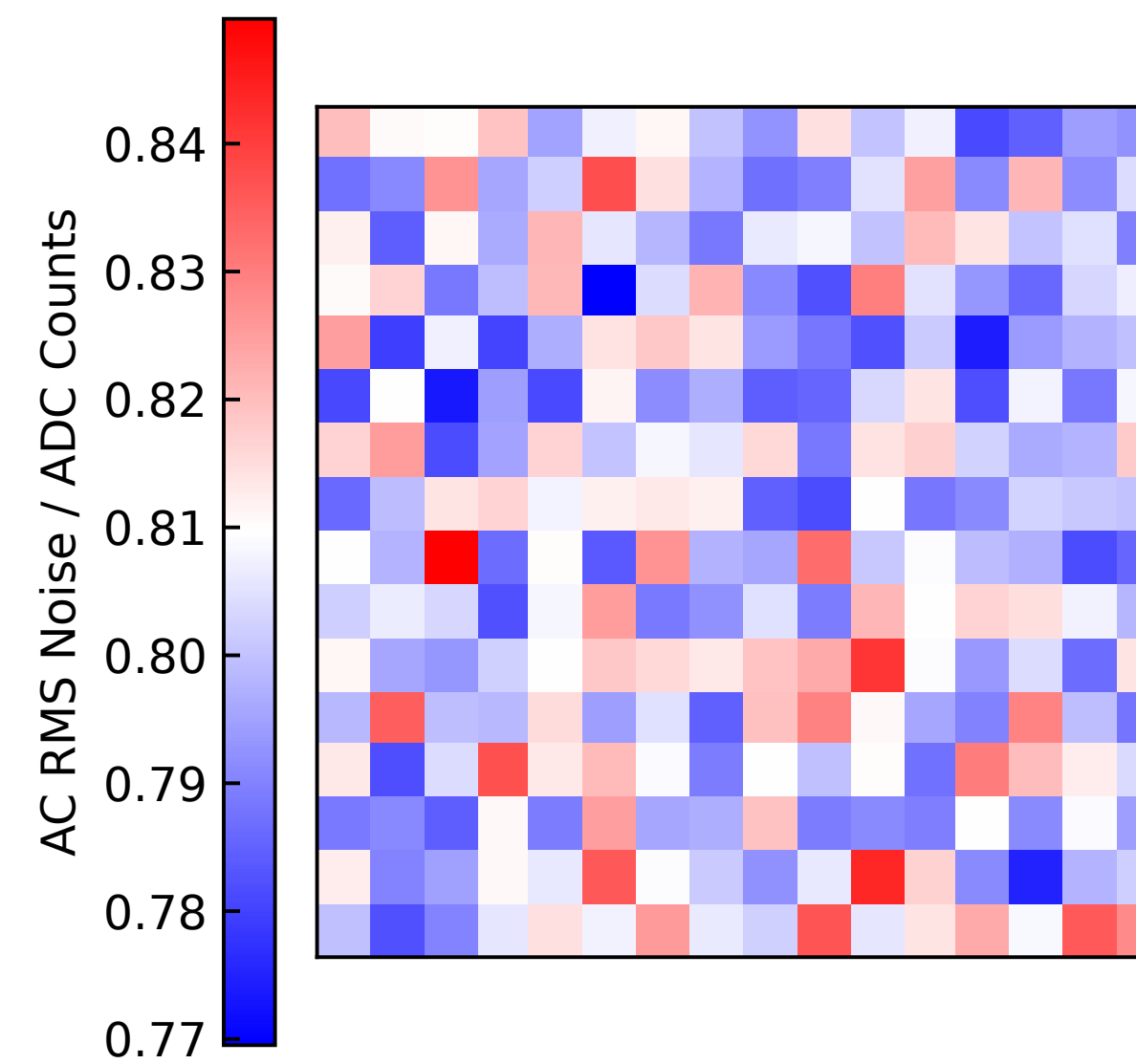
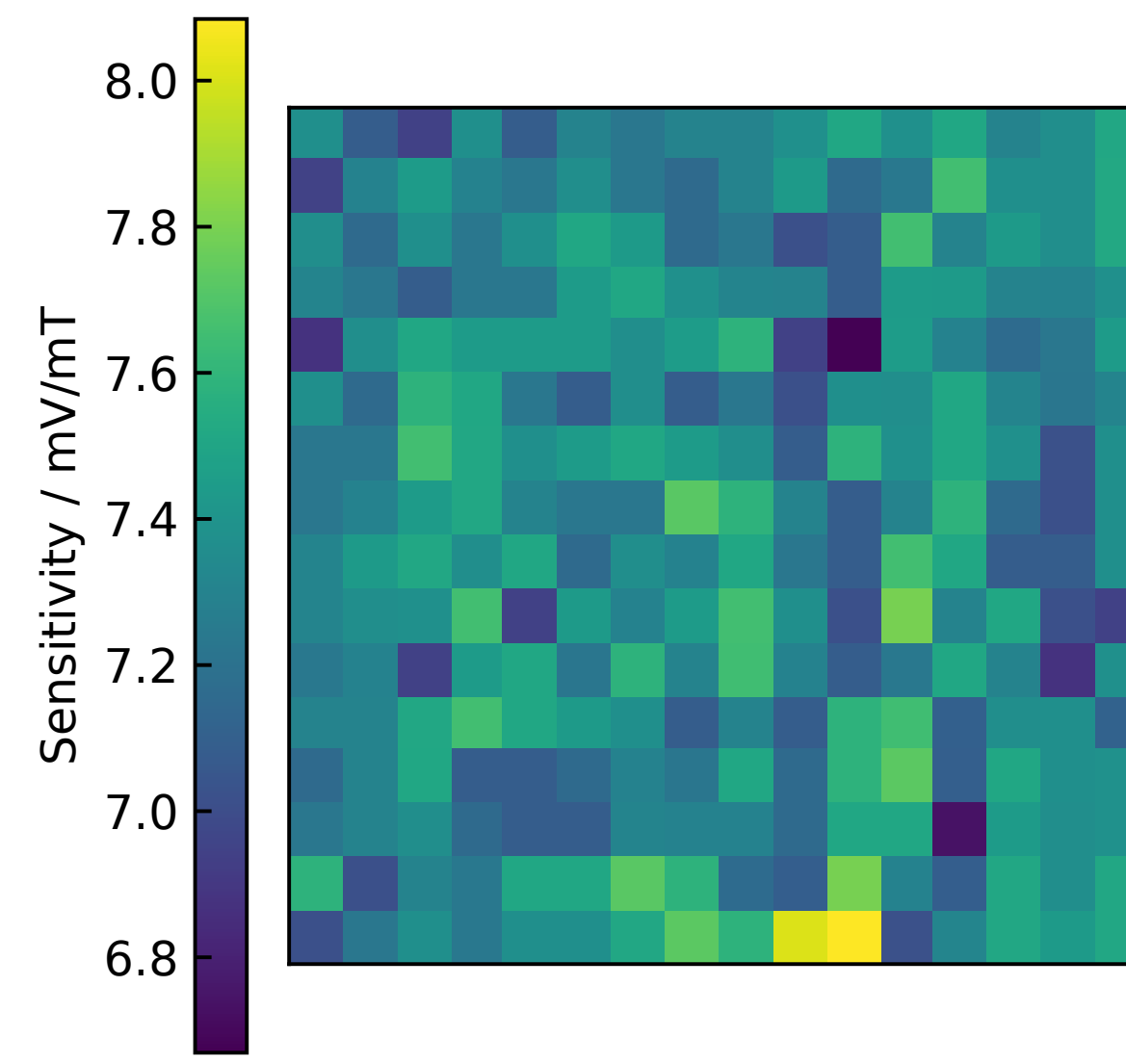
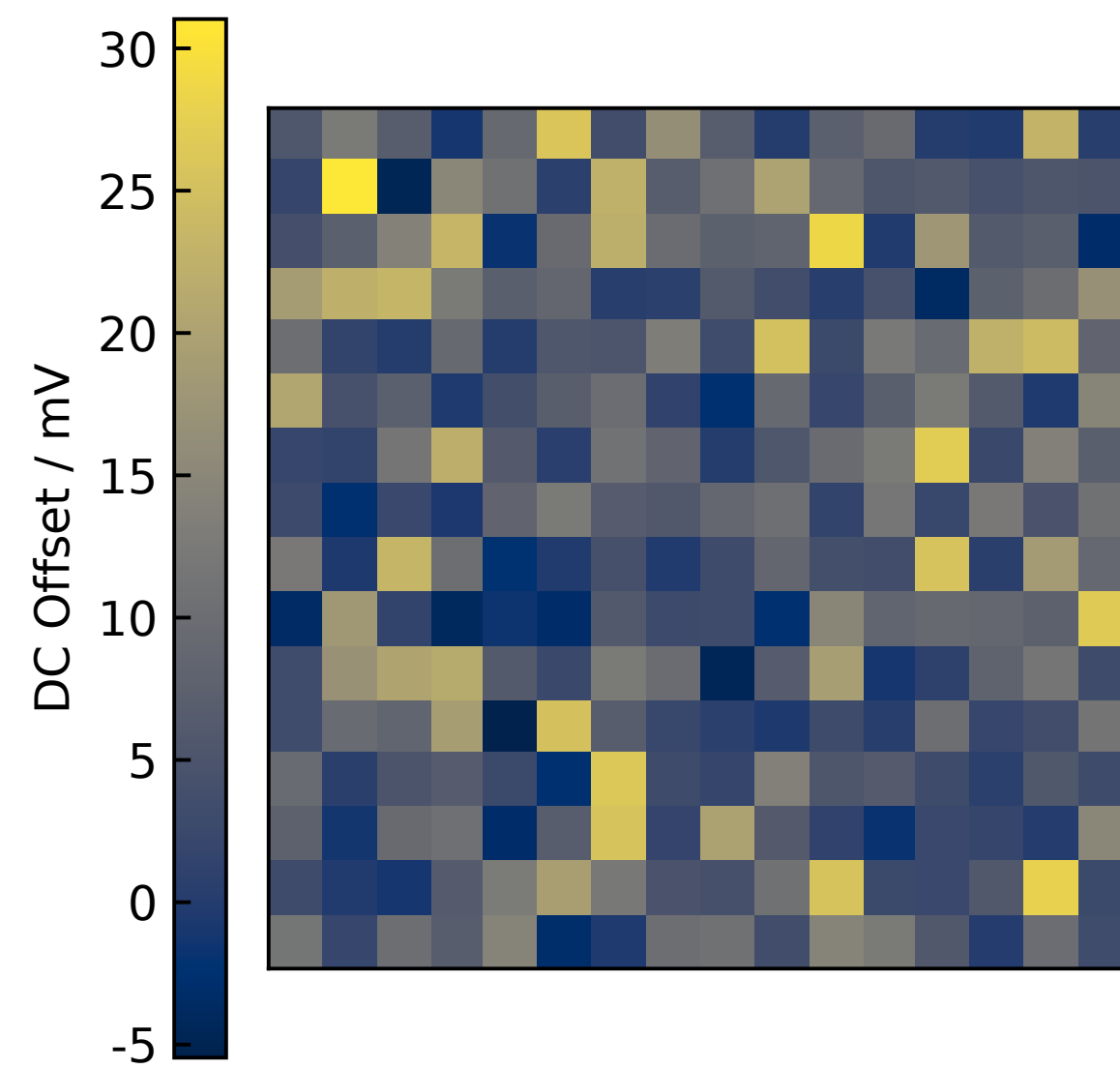
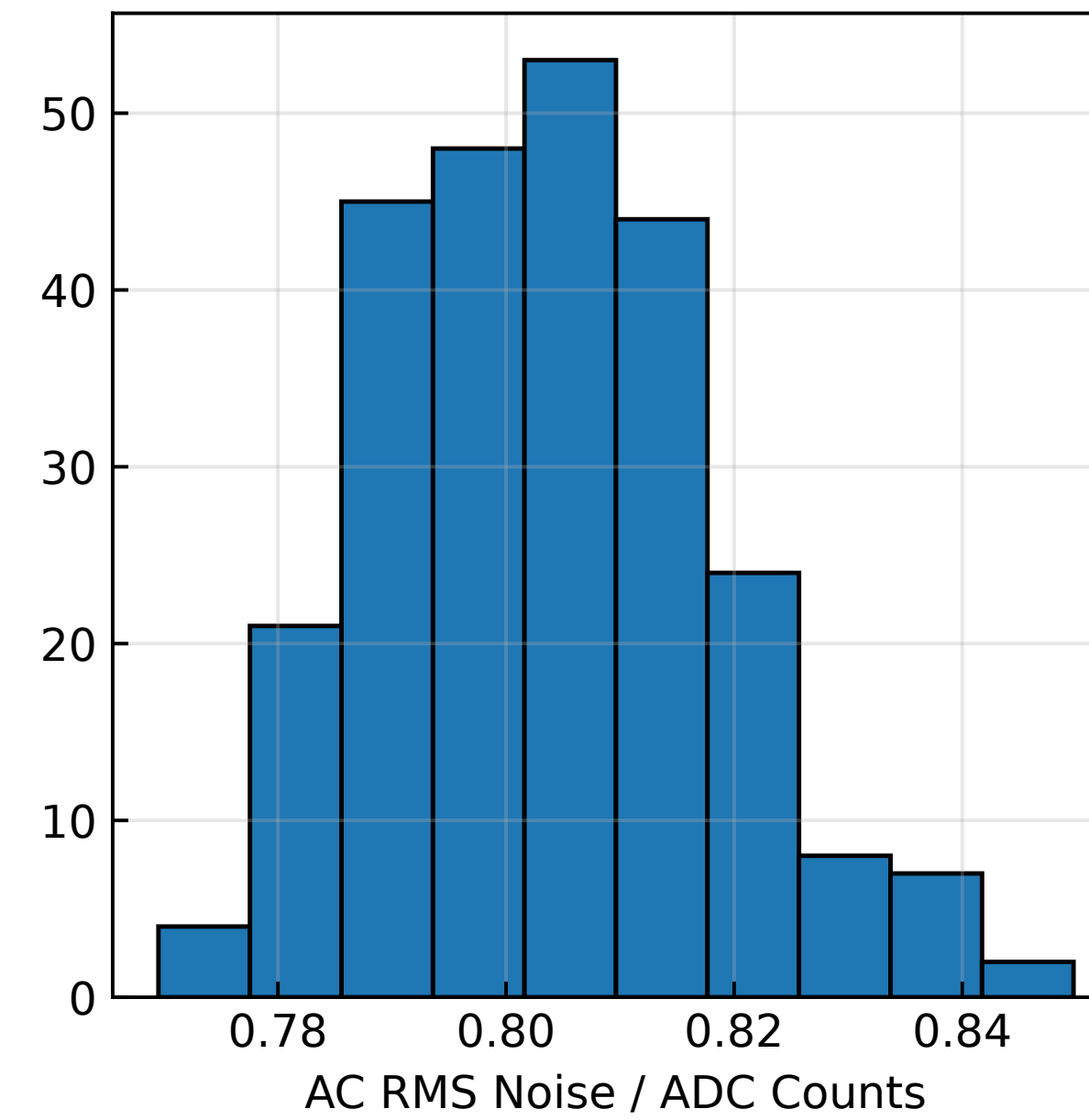
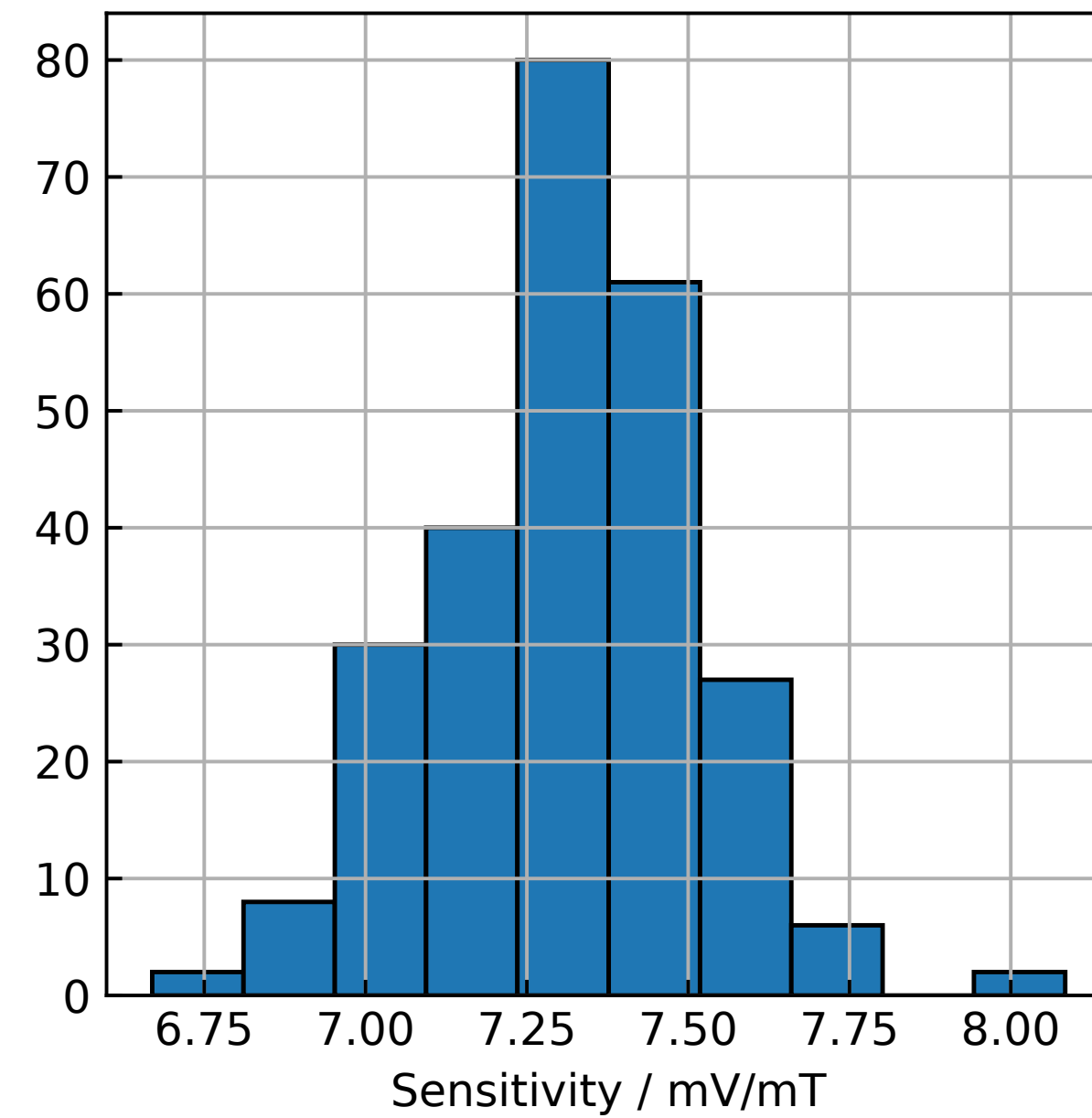
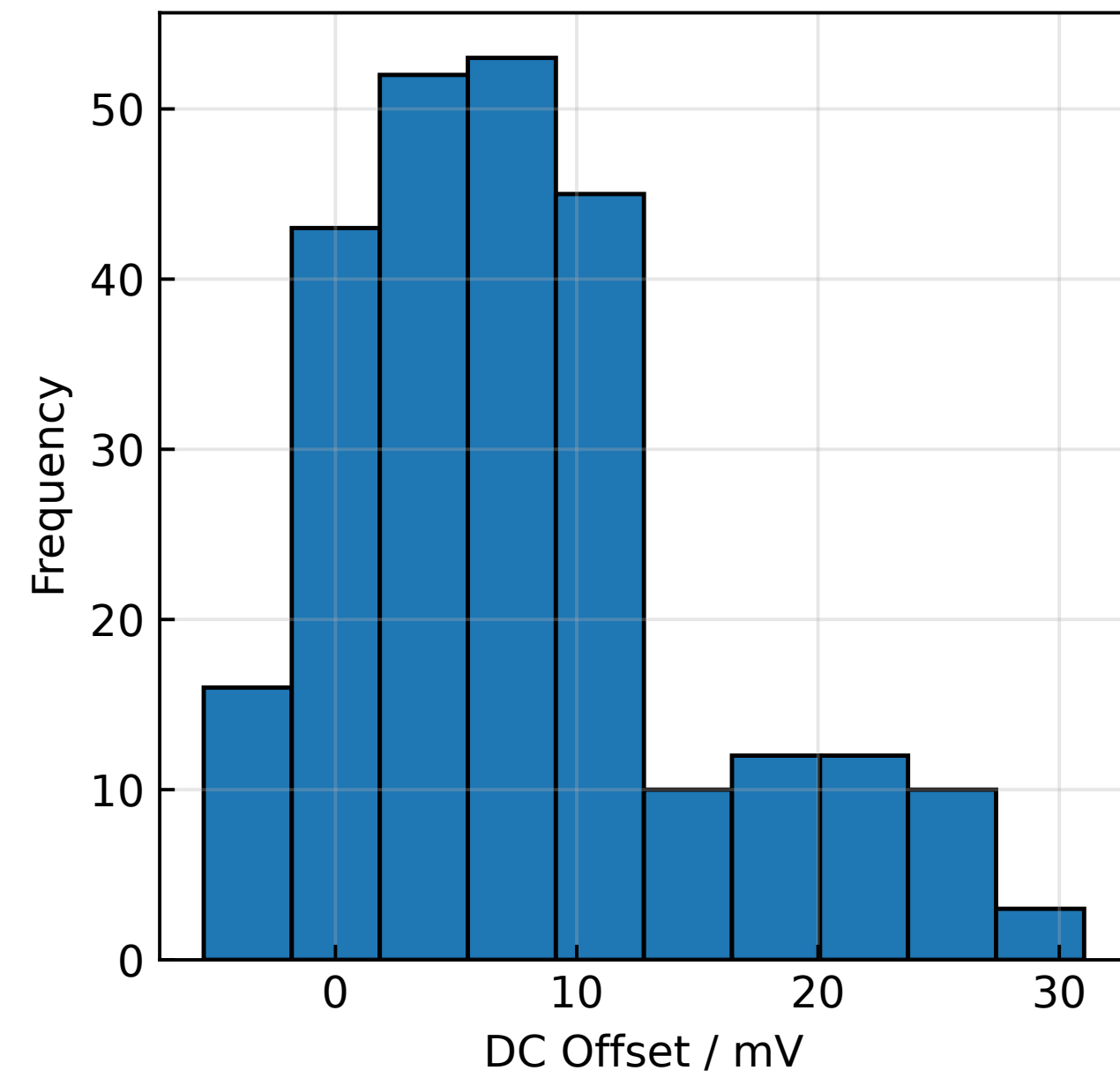
Specifications

Dimensions.....	175×142 mm
Dynamic Range.....	$>\pm 150$ mT
Bandwidth	20 kHz
Sampling	>40 kSPS/sensor
Sensitivity.....	0.32 mT



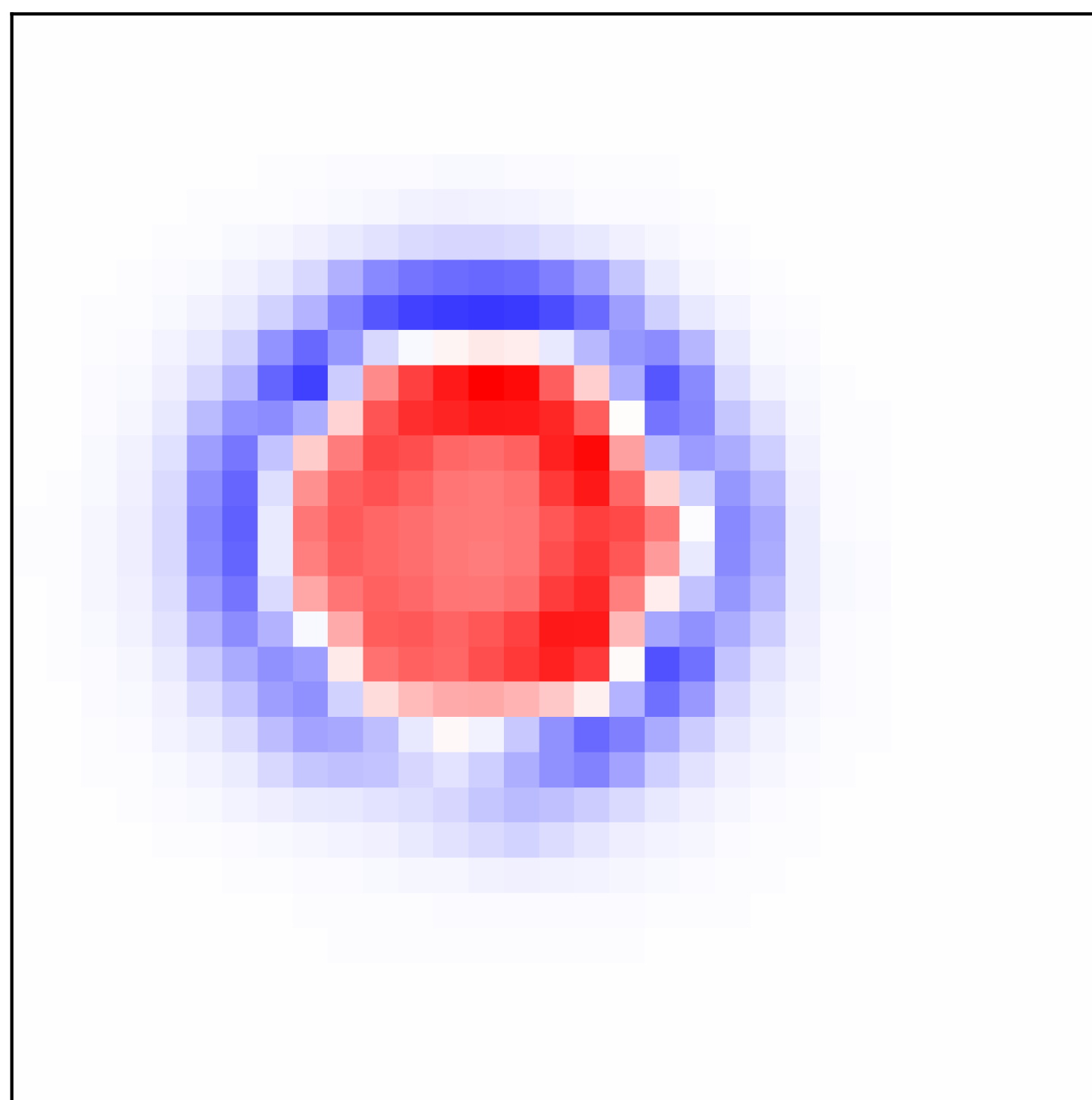
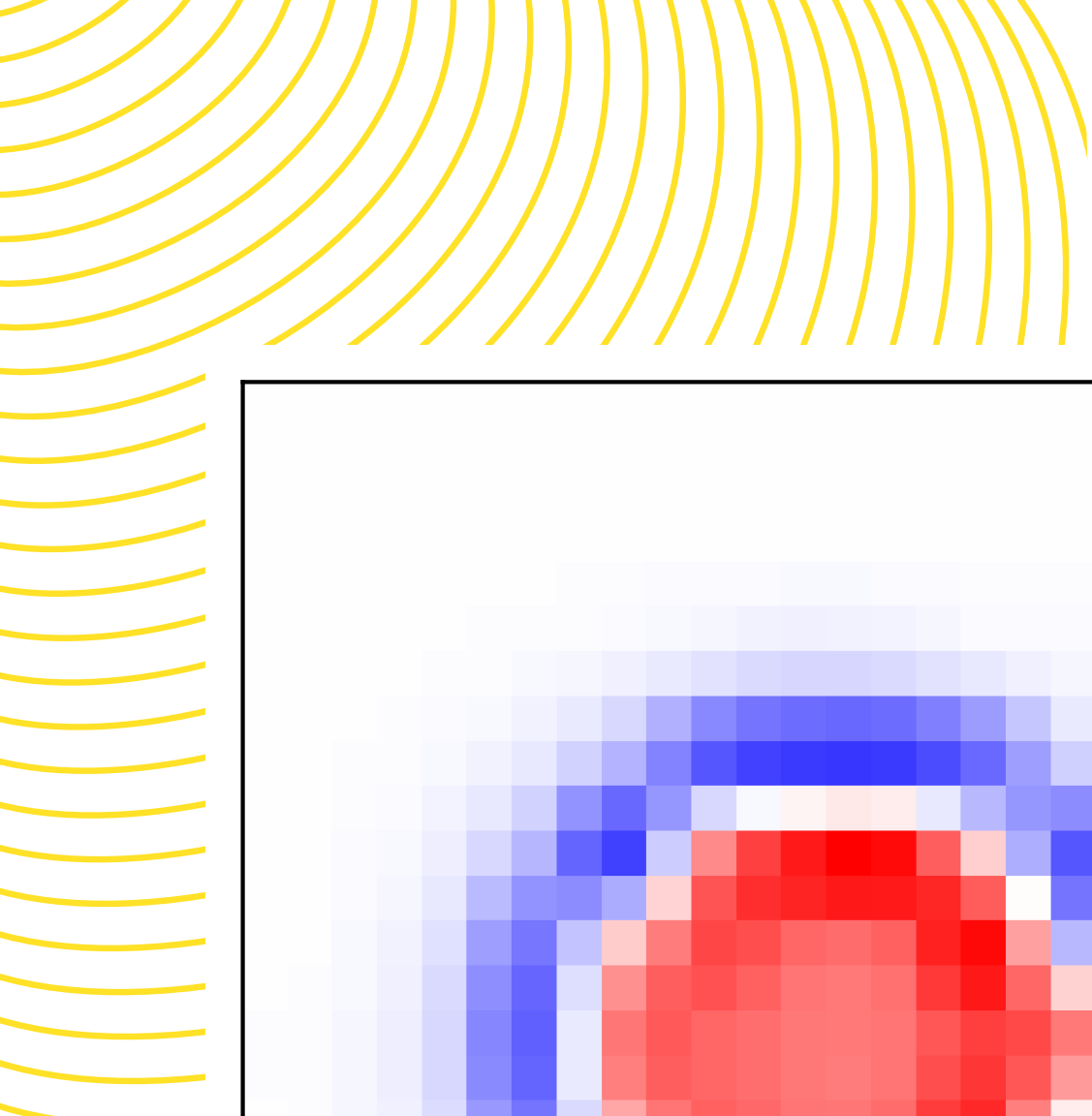
Final Device

Comprehensive fulfilment of all Design Goals

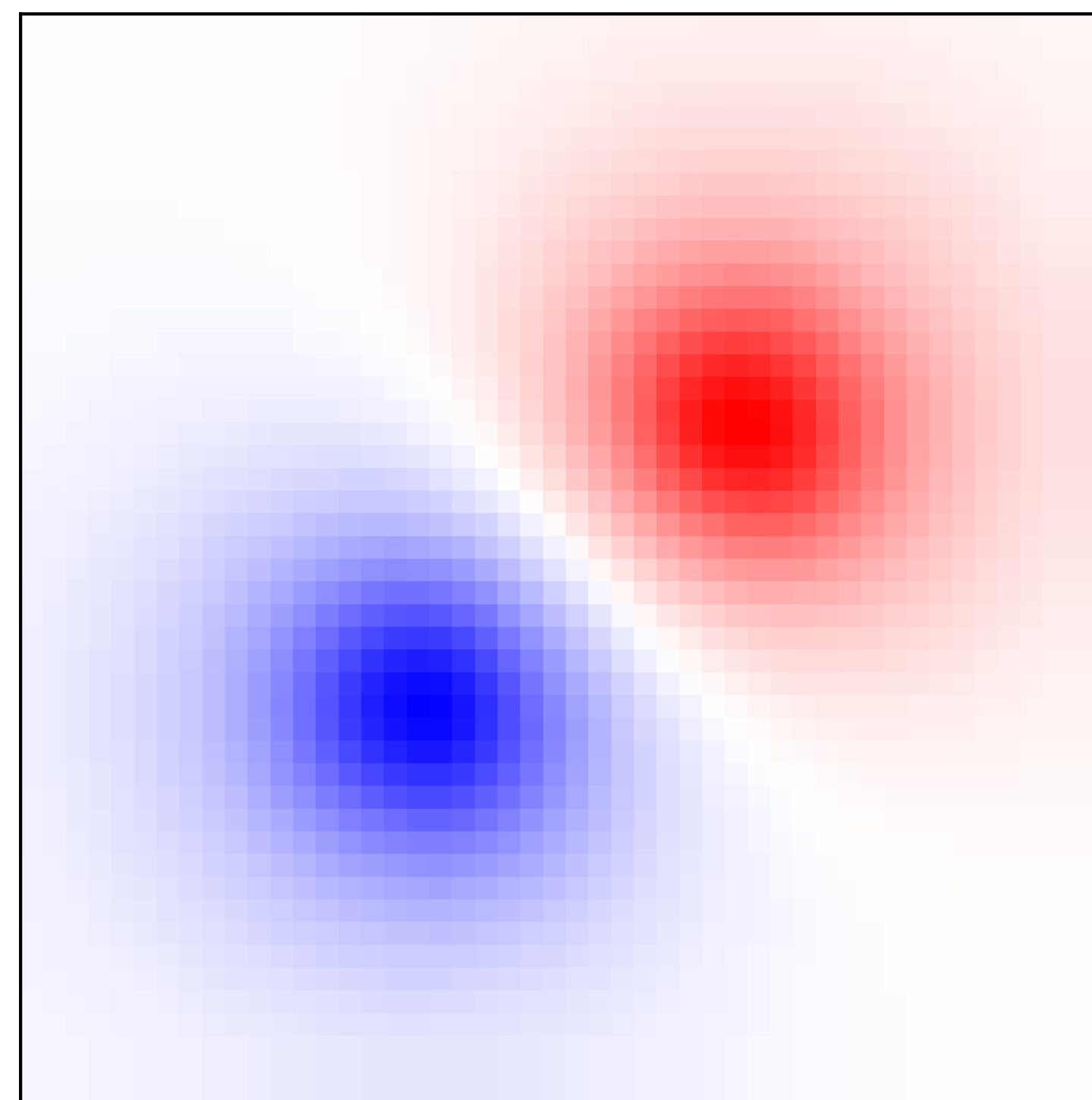
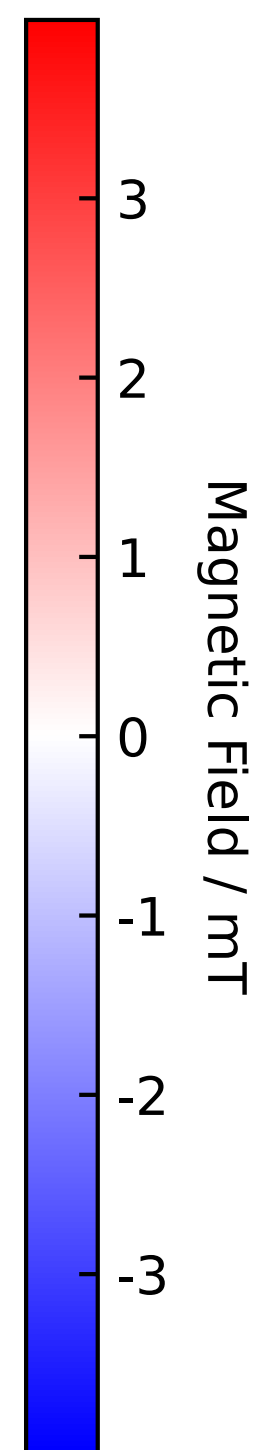


Linear Calibration constants and AC RMS Noise

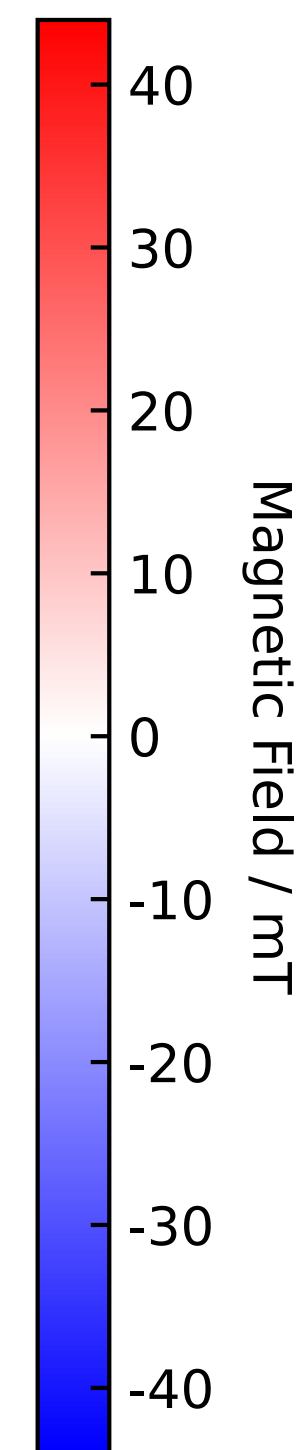
—
Results



MagSafe Charger

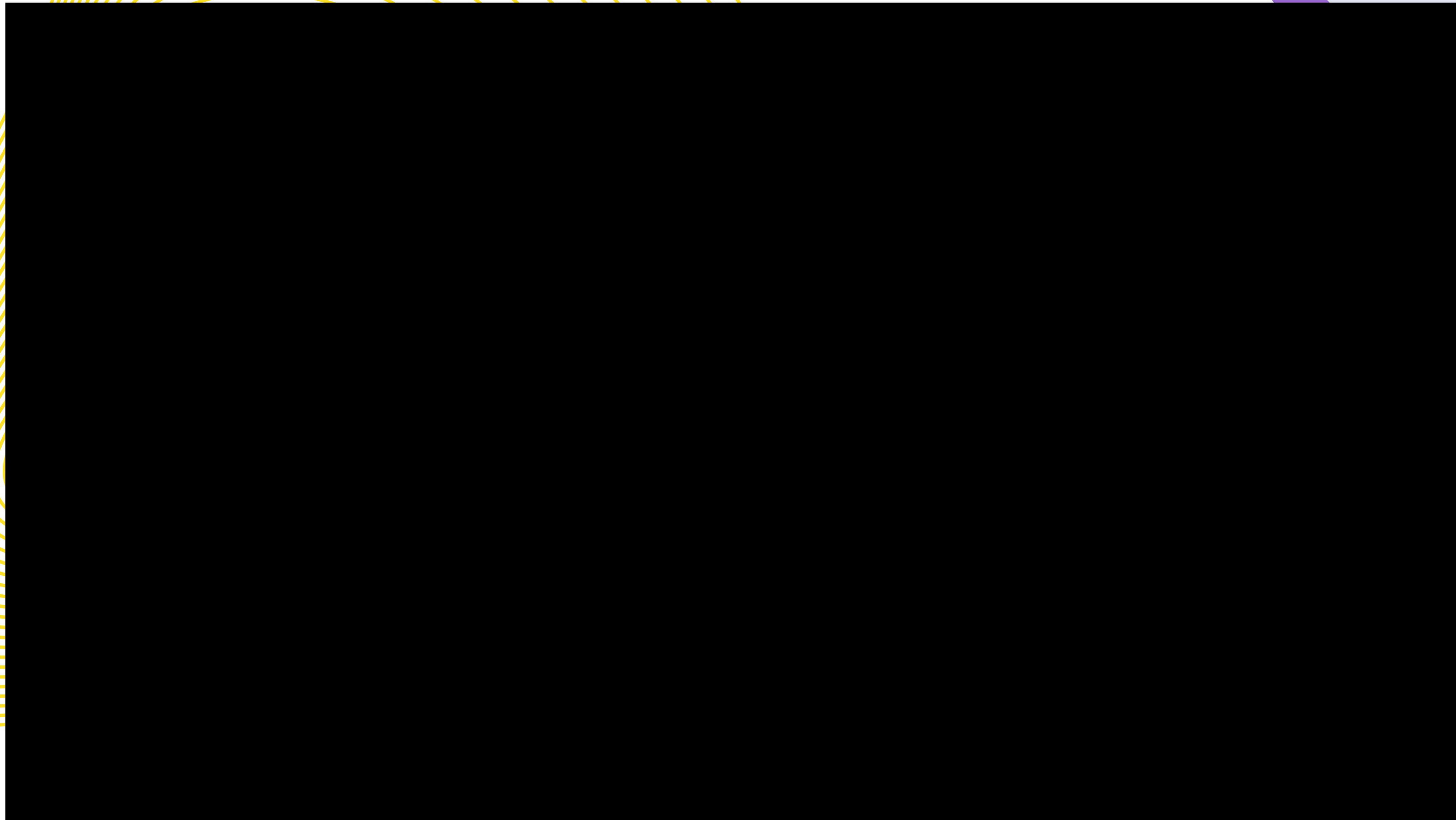


Rare-Earth Bar Magnet



Static Imagery enhanced through **Bicubic Interpolation**

—
Results



— What's Next?

- This work received a **Positive Reception** at the Symposium on Fusion Engineering, in Cambridge, MA; awarded UNSW nuclear innovation centre (UNIC) and Tyree foundation travel grant
- Immediate future priority: development of FPGA frontend to get full 20 kHz bandwidth with >40,000 SPS
- Will be used for SOUTH R&D, 2025–26
- From SOFE: collaboration with SMART
 - Coil alignment for second campaign
 - I will visit them in Seville in October
- Project scope is limited to small devices

The SMART Tokamak⁵

A collaboration between Seville and PPPL

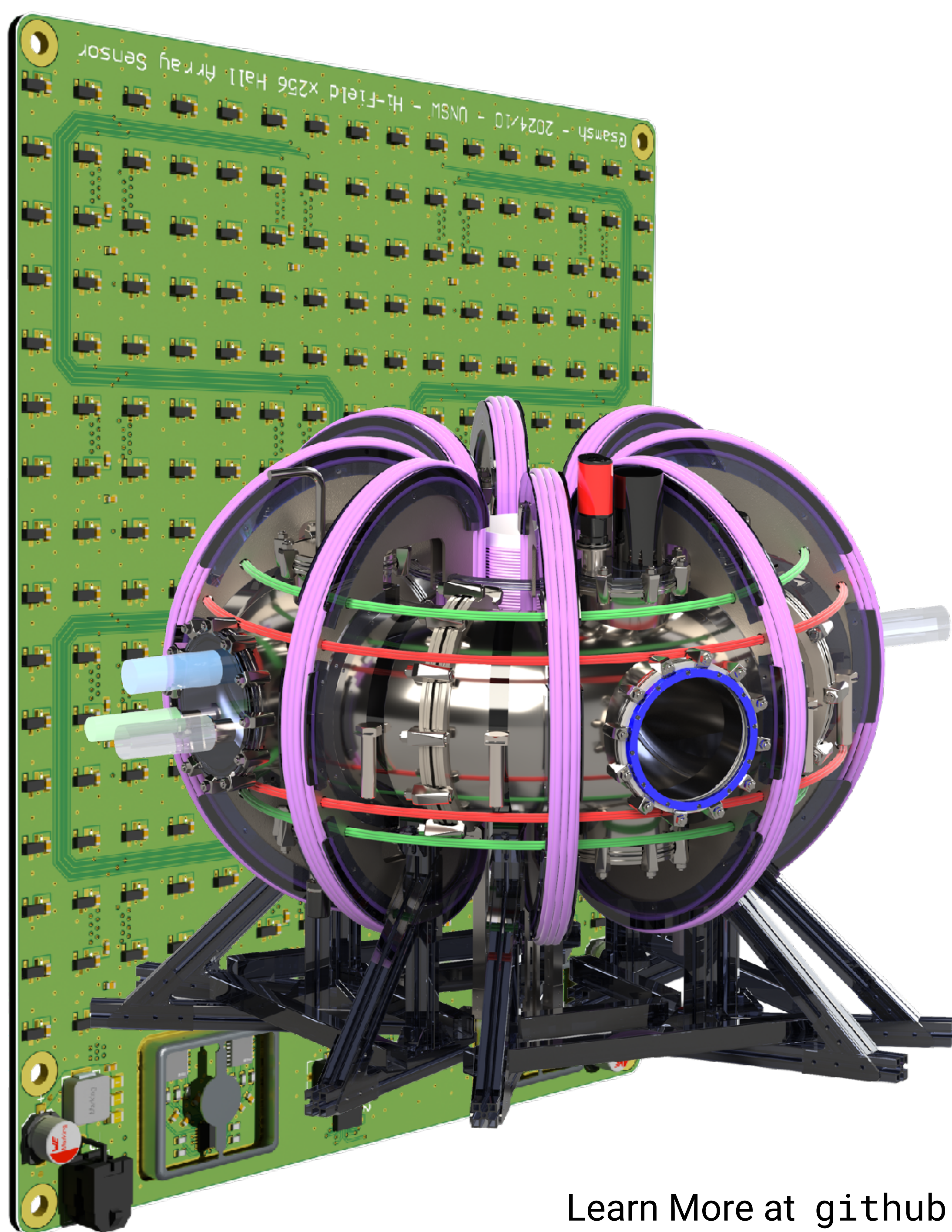


A **magnetic camera** designed to monitor Tokamak toroidal B fields,
designed to address the requirements of the upcoming SOUTH Tokamak:

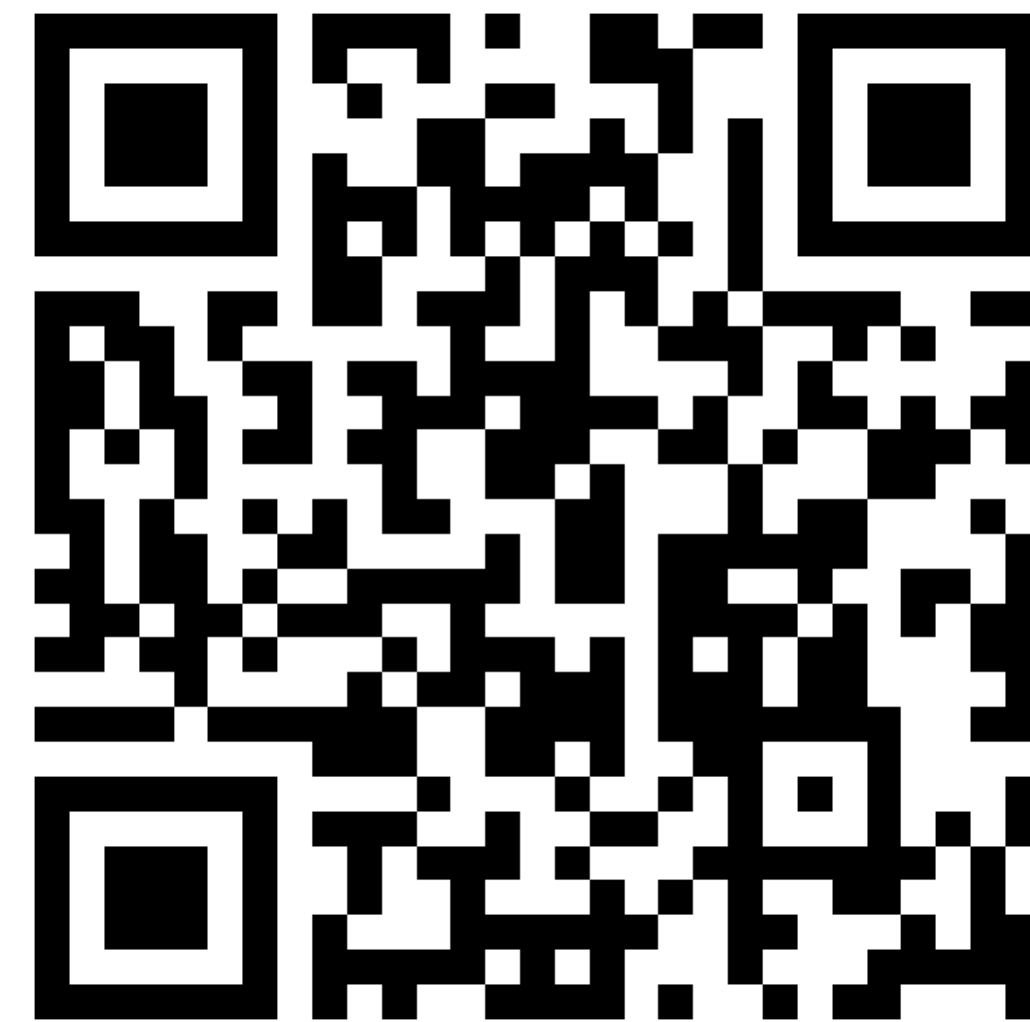
- **20 kHz Bandwidth** achieved through and FPGA and multiple ADCs, or use slower real-time operation with a microcontroller
- **Dynamic range** exceeding of ± 150 mT
- **<1 LSB noise** floor facilitates an effective **sensitivity of ≈ 0.32 mT**

Stay tuned for AtomCraft R&D and SMART commissioning!

Summary



Thank You!



Learn More at github.com/Sam-js2/BETthesis or by scanning the QR code

Bibliography

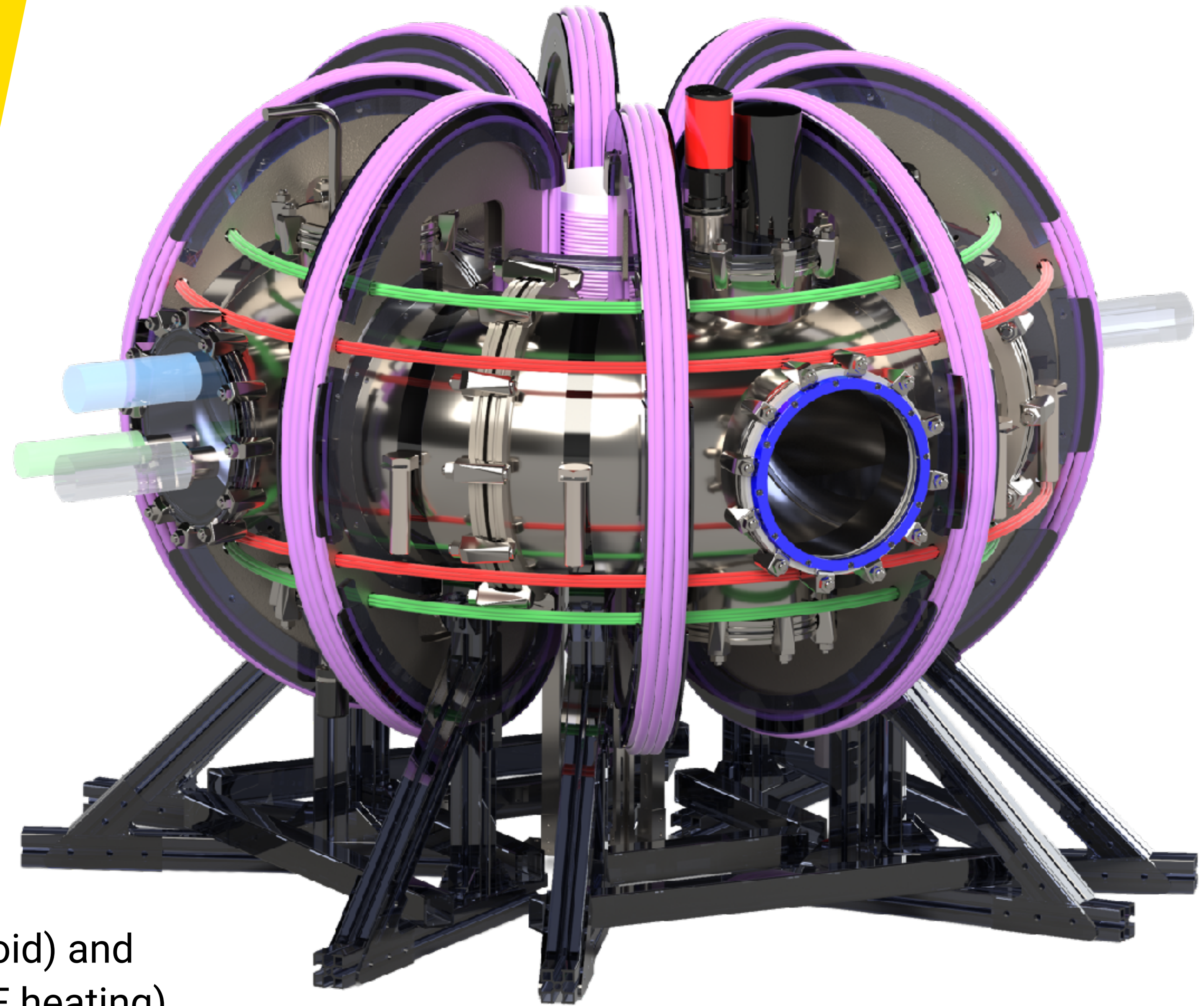
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Backup: SOUTH

Major Radius..... ~0.3 m
Minor Radius 30 cm
Peak Temperature 10 eV*
Plasma Current 3 kA
Toroidal on-axis Field 100 mT
Pulse Duration..... 100 ms

*Thats 120,000 degrees!

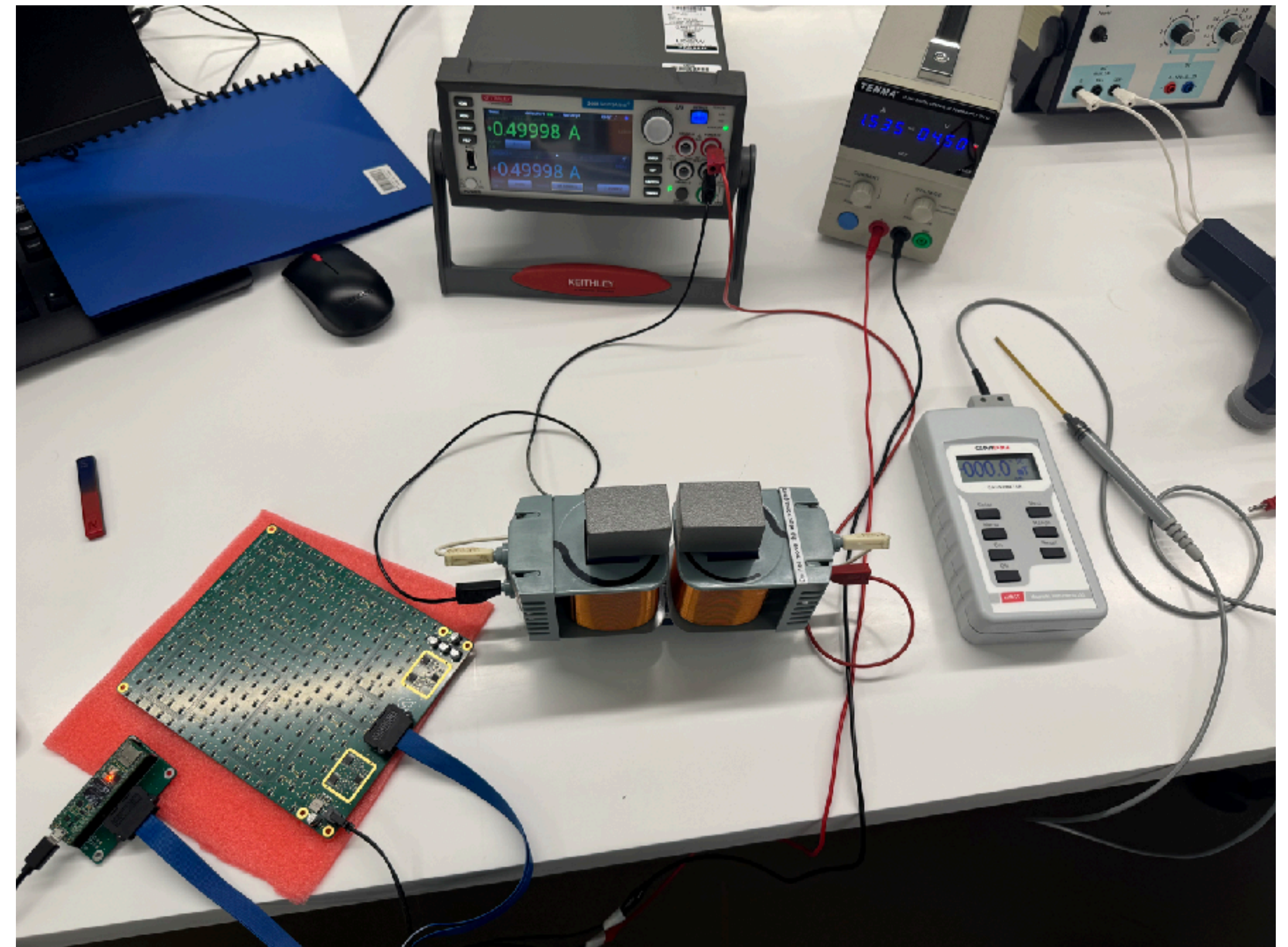
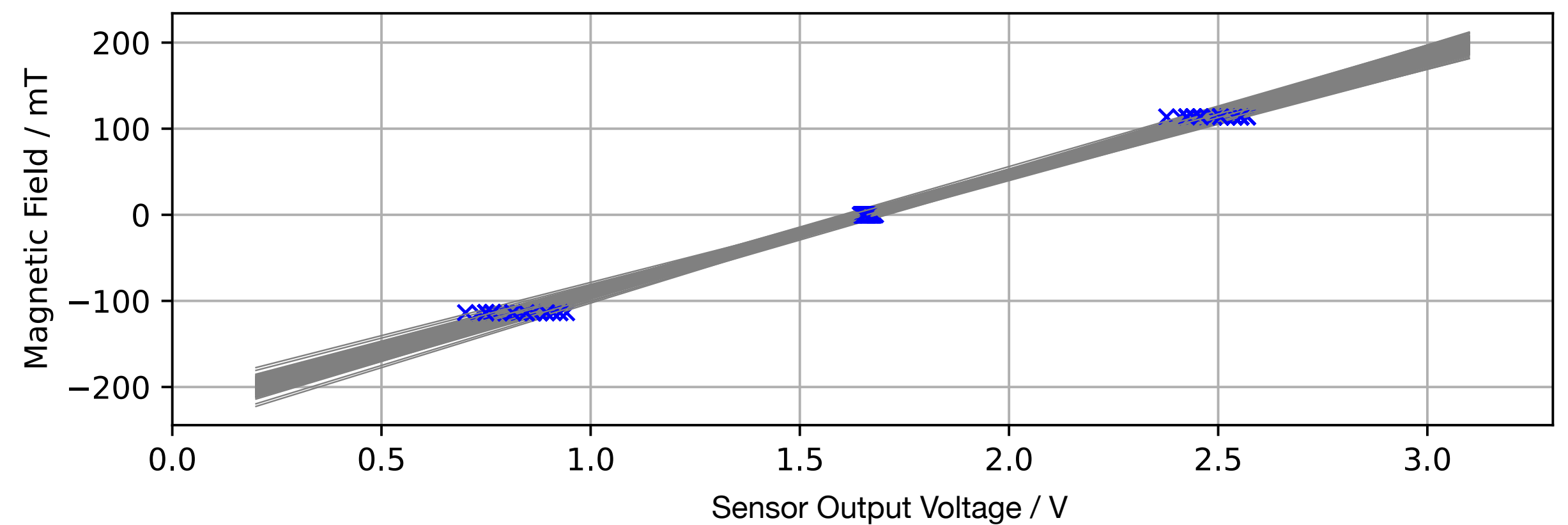
Ionisation with the help of **ohmic heating** (solenoid) and **electron cyclotron resonance heating** (ECRH–RF heating).



Backup: Calibration

Use **ferrite-core electromagnet** powered (extremely precisely) with a **source-meter**:

- ~115 mT field (<1% variations across surface), measured with gauss meter
- Measure aligned and anti aligned (i.e., +/–)
- Gather **lots** of datapoints
- Bin into histogram, take highest frequency bin
- Linear regression on +115/0/–115 mT
- Linearity is already guaranteed by manufacturer, just getting sensitivity



Backup: Digital Workings

